Python Program Outputs

Sample console outputs for problems 52–61

52. Hackathon Team Formation and Ranking

```
Team Beta has an invalid team size

Hackathon Ranking
1 Team Alpha Total: 263 Highest Individual: 90
2 Team Gamma Total: 263 Highest Individual: 89
```

Note: Team Alpha and Team Gamma have equal totals, so Team Alpha ranks first because its highest individual score is greater.

53. Food Waste Monitoring System

```
Food Waste Report
Monday: Prepared=500, Served=450, Leftover=50, Waste=10.00%
  Status: Within acceptable limit
Tuesday: Prepared=600, Served=510, Leftover=90, Waste=15.00%
  Status: EXCEEDS ACCEPTABLE LIMIT
Wednesday: Prepared=550, Served=520, Leftover=30, Waste=5.45%
  Status: Within acceptable limit
Thursday: Prepared=700, Served=600, Leftover=100, Waste=14.29%
  Status: EXCEEDS ACCEPTABLE LIMIT
Friday: Prepared=650, Served=620, Leftover=30, Waste=4.62%
  Status: Within acceptable limit
```

54. Multi-Branch Expense Matrix Analysis

```
Branch A Total = 45000 Average = 11250.0
Branch B Total = 47000 Average = 11750.0
Branch C Total = 45000 Average = 11250.0

Category Totals:
Rent = 36000
Salaries = 44000
Utilities = 33000
Maintenance = 24000

Largest Branch Contributor: Branch B
Largest Category Contributor: Salaries
```

55. Tuple-Based Transaction Audit

```
Duplicate IDs: ['T102']
Invalid Amounts: [('T103', -100, 'WITHDRAW')]

Suspicious Sequences:
('T103', -100, 'WITHDRAW') followed by ('T104', 5000, 'WITHDRAW')
```

56. Student Group Formation with Score Balancing

```
Student Groups

Group 1:
Asha 90
Sneha 68
Total Score: 158

Group 2:
Ravi 85
Rahul 70
Total Score: 155

Group 3:
Kiran 80
Priya 72
Total Score: 152

Group 4:
Meena 78
Arjun 75
Total Score: 153

Score Difference: 6
```

Note: The greedy allocation balances totals but does not guarantee a mathematically optimal grouping for every dataset.

57. Smart Canteen Order and Stock Management

```
Customer 1 Bill = 160
Customer 2 Bill = 300
Customer 3 Bill = 200
Customer 4 : Insufficient stock for Pizza
Customer 5 Bill = 180

Daily Sales: 840

Remaining Stock:
Burger : 8
Pizza : 3
Juice : 10
Sandwich : 5
```

58. Library Borrowing and Fine Reconciliation

```
Library Fine Report
B101 Overdue Days: 0 Fine: 0 Status: ON TIME
B102 Overdue Days: 3 Fine: 15 Status: OVERDUE
B103 Overdue Days: 7 Fine: 35 Status: OVERDUE
B104 Overdue Days: 0 Fine: 0 Status: ON TIME
B105 Overdue Days: 5 Fine: 25 Status: OVERDUE

Total Fine: 75
```

Note: The supplied code calculates overdue days from borrowed_days and the seven-day limit; it does not use the separate delay value in each tuple.

59. Campus Transport Allocation

```
Campus Transport Allocation
Route A
  Passenger Demand: 95
  Buses Required: 2
  Available Capacity: 100
  Unserved Passengers: 0
Route B
  Passenger Demand: 140
  Buses Required: 3
  Available Capacity: 150
  Unserved Passengers: 0
Route C
  Passenger Demand: 72
  Buses Required: 2
  Available Capacity: 100
  Unserved Passengers: 0
Route D
  Passenger Demand: 210
  Buses Required: 5
  Available Capacity: 250
  Unserved Passengers: 0

Total Buses Required: 12
```

60. Manufacturing Quality Control Matrix

```
Manufacturing Quality Control Report

Product A
Batch 1: Production=1000, Defects=20, Defect%=2.00%, ACCEPTABLE
Batch 2: Production=1200, Defects=50, Defect%=4.17%, QUALITY VIOLATION
Batch 3: Production=1100, Defects=15, Defect%=1.36%, ACCEPTABLE

Product B
Batch 1: Production=800, Defects=40, Defect%=5.00%, QUALITY VIOLATION
Batch 2: Production=900, Defects=30, Defect%=3.33%, ACCEPTABLE
Batch 3: Production=1000, Defects=25, Defect%=2.50%, ACCEPTABLE

Product C
Batch 1: Production=1500, Defects=60, Defect%=4.00%, ACCEPTABLE
Batch 2: Production=1400, Defects=35, Defect%=2.50%, ACCEPTABLE
Batch 3: Production=1600, Defects=80, Defect%=5.00%, QUALITY VIOLATION
```

Note: The threshold is 4%; only defect percentages strictly greater than 4% are flagged.

61. Nested List Examination Seating Validation

```
Examination Seating Validation

Duplicate Roll Numbers:
[102]

Reserved Seat Violations:
None

Empty Seats:
Position: (1, 1)

Invalid Student Allocations:
Position: (3, 2) Roll Number: 999

Total Validation Errors: 3
Seating arrangement contains ERRORS
```

Note: The reserved position (1, 1) contains None in the sample data, so it is counted as an empty seat, not as a reserved-seat violation.